Majid Farhadi

farhadi@ualberta.ca Edmonton, Alberta, Canada majidfarhadi78 majid-farhadi

Education

University of Alberta, Computer Engineering — Edmonton, Canada, 2023 – 2026

Isfahan University of Technology, Computer Engineering — Isfahan, Iran, 2018 – 2023

Experience

Very Large Scale Integration (VLSI) Research Lab, University of Alberta, Graduate Research Assistant — Edmonton, Canada, 2023 – 2026, 3 years

- Designed and implemented AI architectures, including training and inference circuits, on FPGAs.
- Integrated and synchronized IP core modules with edge detector 1-stage synchronizers.
- Wrote testbench modules for behavioral, functional, and timing simulation and verification of RTL designs.
- Defined implementation constraints including clock period to meet timing requirements.
- Optimized dynamic power by minimizing switching activity through stop conditions.
- Fine-tuned distance functions in clustering algorithms and derived gradients in backpropagation algorithms.

Structured Machine Learning Research Lab, Simon Fraser University, Graduate Research Assistant — Burnaby, Canada, 2023 – 2024, 1 year

- Counted motifs in a graph database by matrix multiplication, suitable for ML applications.

Embedded Systems and Networks Research Lab, Isfahan University of Technology, Undergraduate Research Assistant — Isfahan, Iran, 2021 – 2023, 2 years

- Designed an interface for a hardware accelerator for the time-slotted channel hopping (TSCH) protocol.
- Stacked on top of an IEEE 802.15.4 transceiver used in wireless sensor networks in IoT applications.
- Deployed on an Altera DE1-SoC board, featuring a Cyclone FPGA and a dual-core Cortex-A9 ARM.
- Integrated the ARM processors and FPGA fabric through a lightweight master-slave AXI bridge.
- Enabled full-duplex read/write communication between FPGA and ARM processor.
- Compiled C programs for the target ARM architecture using makefiles.
- Accessed the FPGA from C programs via virtual addresses provided by memory mappings.
- Offloaded computation from the processor and exploited the FPGA's high throughput.

Kian-Pardaz Naghsh-e-Jahan Engineering Company, Intern — Isfahan, Iran, 2021 – 2021, 1 year

- Designed a PCB and implemented a web client for a smart thermostat on an ESP8266-12E module.

Publications

SCFNNs: Stochastic Computing Fuzzy Neural Networks
Majid Farhadi, Witold Pedrycz, Jie Han
ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11360989

Rule-Enhanced Graph Learning
Ali Khazraee, Abdolreza Mirzaei, Majid Farhadi, Parmis Nadaf, Kiarash Zahirnia, Mohammad Salameh, Kevin Cannons, Richard Mar, Mingyi Wu, Oliver Schulte
openreview.net/forum?id=m02qzfjlHA&referrer=%5Bthe%20profile%20of%20Kiarash%20Zahirnia%5D(%2Fprofile%3Fid%3D~Kiarash

1BCMs: 1-Bit Cognitive Maps
Majid Farhadi, Witold Pedrycz, Jie Han

A Memory-Efficient and Low-Complexity Probabilistic Sigmoidal Approximator
Wenhao Lu, Majid Farhadi, Xingjian Gao, Chris Lynch, Jie Han

Enhancing Graph Generation With First-Order Logic Rules
Ali Khazraee, Abdolreza Mirzaei, Majid Farhadi, Parmis Nadaf, Kiarash Zahirnia, Mohammad Salameh, Kevin Cannons, Richard Mar, Mingyi Wu, Oliver Schulte